FIT5120 Industry Experience Studio Project
Data Management Plan

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1. Datasets Overview

The PlantX data ecosystem integrates a range of **open-source, API-based, and curated datasets** into a unified relational database designed to support data-driven features across the application lifecycle. These datasets span general plant information, species-specific care requirements, spatial distribution patterns, biodiversity indicators, climate variables, and disease classification images.

By consolidating these diverse datasets into a single linked system, PlantX enables a broad range of user-facing capabilities, including:

- **Search and Filtering:** Users can query plants based on specific biological and horticultural attributes.
- **Personalised Care Guidance:** Context-aware care recommendations are tailored to environmental conditions, species characteristics, and geographic context.
- **Image-Based Recognition:** One-shot image retrieval and disease classification allow users to identify plants and potential health issues from photos.
- **Biodiversity and Climate Insights:** Long-term ecological trends, threatened species monitoring, and climate-forecast-driven recommendations support sustainable gardening and conservation awareness.

This multi-source data integration underpins the platform's ability to deliver both consumer-focused functionalities (e.g., personalised gardening recommendations) and research-aligned insights (e.g., biodiversity forecasting and conservation mapping), directly supporting UN SDG 13 (Climate Action).

1.1. Data Sources

Names	**Physical access**	**Frequency of source updates**	**Frequency of iteration system updates**	**Granularity**	**Copyright/licensing details**
Prenual Plant API: Species Details	API (JSON)	Periodic	On demand	High	Prenual API Terms and Use
Prenual Plant API: Care Guides	API (JSON)	Periodic	On demand	High	Prenual API Terms and Use
Prenual Plant API: Distribution/Hardiness Map	API (HTML)	Periodic	On demand	Medium	Prenual API Terms and Use
City of Melbourne: Threatened Plant Living	CSV	Periodic	Annually	High	Melbourne Open Data licence

Collection Plan					
City of Melbourne: Urban Forest	CSV	Periodic	Annually	High	Melbourne Open Data licence
Open-Meteo: Historical & Forecast Weather	API (JSON)	Real-time	On demand	Medium	Open-Meteo terms
Threatened Species (TSX)	CSV	Annual	On demand	Yearly index	TSX terms
Kaggle - Plant Disease Dataset	Files (JPG)	Static	As needed	High (labelled image)	Kaggle dataset terms
Australian Garden Club Directory	CSV	Ad-hoc	As needed	Organisation level	Public listings
ASGS State/Territory Digital Boundaries	SHP / GeoJSON	Periodic (ABS releases)	As needed	Region-level	ABS license (open)

1.1.1 Perenual – Species Details

Provides detailed biological and horticultural data for thousands of plant species, including taxonomy, physical characteristics, growth rate, life cycle, and environmental preferences. This dataset forms the foundation of plant search and filtering functions and underpins the plant profile pages.

1.1.2 Perenual – Care Guides

Delivers care instructions tailored to each species, covering key parameters such as watering frequency, sunlight exposure, soil composition, pruning schedules, and propagation methods. These records are central to the dynamic care guide pages and personalised recommendations.

1.1.3 Perenual – Distribution/Hardiness Map

Provides spatial distribution and climatic hardiness data, indicating the geographic regions where each species thrives. This information enhances recommendation features and informs users about the influence of location and climate on plant viability.

1.1.4 City of Melbourne – Threatened Plant Living Collection Plan

Contains conservation status, ecological requirements, and care details for native threatened plant species in Victoria. This dataset enables biodiversity-focused insights and allows the system to differentiate between general and threatened species, directly supporting the conservation and educational objectives of the project/

1.1.5 City of Melbourne – Urban Forest
Provides geospatial data on urban tree populations, including species, location, physical dimensions, and life expectancy. It supports spatial visualisation features and aids in identifying underrepresented species for more sustainable urban planting strategies.

1.1.6 Open Meteo – Weather Forecast
Supplies real-time and forecasted weather data such as temperature, rainfall, and solar radiation. This data is used to dynamically match plants to short-term climate conditions, enabling predictive and adaptive recommendations.

1.1.7 TSX – Threatened Species
Provides time-series data on species population trends in Victoria, normalised to a baseline year (1985). These data underpin long-term biodiversity trend analysis, foresight modelling, and predictive conservation features

1.1.8 Kaggle – Plant Disease Dataset
Contains labelled images of plant diseases across multiple species. When mapped to Perenual's taxonomy, it serves as the training dataset for the disease recognition model and supports diagnostic image classification features.

1.1.9. Garden Clubs – Australian Garden Club Directory
Curated organisational metadata for Australian garden clubs, including club name, website link, meeting cadence (day/week/hour), contact, and location/state. This dataset powers community discovery and engagement features, enabling users to find relevant clubs by schedule and region and to connect platform insights with local gardening communities.

1.1.10. ABS – ASGS State/Territory Digital Boundaries
Provides authoritative state and territory boundary geometries for Australia (stored as MULTIPOLYGON, SRID 7844). These shapes enable choropleth mapping, regional aggregation, and spatial joins with biodiversity and climate outputs (e.g., enriched TSX trends), ensuring cartographic accuracy and consistent regional analysis across the system.

2. Data Usage

Our datasets form the foundation of the system's functionality, enabling a range of features across multiple iterations, from fundamental plant search and care guidance to advanced forecasting, spatial analysis, and community engagement. Each dataset contributes to specific application features and user stories as outlined below.

Prenual (Species Details, Care Guides, Images, Distribution Maps)

- **Search and Filter (Iteration 1 – Epic 2, US 2.1):**
  Core attributes such as watering needs, sunlight exposure, growth rate, edibility, medicinal properties, toxicity, plant cycle, and growth rate power the plant search and filtering engine.
- **Plant Profile & Care Guidance (Iteration 1 – Epic 2, US 2.2):**
  Detailed care information (watering, sunlight, pruning, soil) enhances plant detail pages and tutorial content, improving user understanding and engagement.
- **Image-Based Search (Iteration 1 – Epic 2, US 2.1):**
  Thumbnail images provide essential training data for the image retrieval model, enabling one-shot image-based plant identification.
- **Geographic Context (Iteration 1 – Epic 2, US 2.2):**
  Distribution and hardiness maps integrate environmental context into plant profiles, showing how regional climate impacts care needs and growth suitability.

City of Melbourne (Threatened Plant Living Collection Plan)

- **Threatened Plant Classification (Iteration 1 – Epic 2):**
  Adds classification layers to distinguish threatened species from general plant data, improving plant detail pages and enriching conservation-focused content.
- **Local Care Adaptation:**
  Provides native-specific care guidelines and provenance data, contextualising advice for Australian climates and conservation efforts.

Plant Disease Data (Perenual + Kaggle)

- **Disease Search and Diagnosis (Iteration 2 – Epic 3, US 3.1–3.3):**
  Disease metadata (scientific/common names, hosts, descriptions, and solutions) powers the disease search and knowledge base.
- **Image Recognition Pipeline (Iteration 2 – Epic 3):**
  Regular and thumbnail images are used to train disease identification models.
- **Cross-Reference Mapping:**
  The linkage between Kaggle and Perenual datasets standardises disease taxonomy, enabling accurate classification and interpretation across different data sources

Urban Forest and Weather Data

- **Urban Forest Spatial Insights (Iteration 2 – Epic 5):**
  Tree distribution and spatial attributes power interactive mapping features and spatial queries, helping users visualise and analyse Melbourne's urban forest coverage.
- **Weather-Linked Plant Recommendation (Iteration 2 – Epic 4):**
  Daily and forecasted weather conditions (temperature, precipitation, sunlight) are matched with plant tolerance data to recommend suitable species for upcoming climatic conditions

Threatened Species Index + Climate Data (TSX + Open-Meteo)

- **Biodiversity Trend Analysis (Iteration 3 – Epic 7):**
  Annual TSX data enriched with climate variables (temperature, rainfall, radiation) supports time-series modelling to identify long-term biodiversity trends.
- **Predictive Forecasting (Iteration 3 – Epic 7):**
  ARIMA/SARIMAX models leverage these enriched datasets to forecast future population trends and climate trajectories, assisting policymakers and users in planning conservation actions

Species Monitoring and Spatial Layers

- **Species-Level Insights (Iteration 3 – Epic 6 & 7):**
  Detailed species-level monitoring data underpins drill-down analytics, allowing users to explore conservation status and distribution trends at a granular level.
- **Geospatial Analysis (Iteration 3 – Epic 7):**
  State boundary shapefiles enable spatial joins and regional visualisation of biodiversity and climate data

Community Engagement (Garden Club Data)

- **Community Discovery (Iteration 3 – Epic 8):**
  The Australian Garden Club dataset enables users to connect with local gardening communities, find clubs based on meeting schedules and location, and participate in events , extending the platform beyond information delivery to active engagement.

3. Data Preparation

This section documents the end-to-end preparation of external datasets into analysis-ready and application-ready tables. We describe extraction, transformation, normalisation standards, quality controls, and outputs. Detailed implementation scripts are maintained in the project repository; this report focuses on data decisions and reproducibility.

For more detailed information, refer to: <Data List>.

3.1. Data Cleaning and Wrangling

3.1.1. Perenual (Species Details, Care Guides, Distribution/Hardiness Map, Images)

Purpose

Perenual is the primary source for general plant identity, traits, care guidance, distribution maps, and images. It underpins search/filter (Iteration 1 – Epic 2), plant detail pages, care tutorials, and feeds the image retrieval model (Iteration 1 ML).

Extraction

Data were retrieved via Perenual REST endpoints (species details; care guides; hardiness map links). Responses were saved as one JSON per species (IDs 1–3000) and companion JSON for care guides. Image URLs (regular and thumbnail) were recorded for later downloading. Keys are managed via environment variables; raw pulls are resume-safe and logged.

Transformation & Normalisation

- **Identity unification:** one row per species; `plant_id = general_plant_id` (1–3000). Threatened plants are appended later from City of Melbourne with `plant_id = threatened_plant_id` (3001–3200) to avoid collisions
- **Boolean flags:** derived from source attributes and normalised to TRUE/FALSE/NULL for:
    - edibility (`edible_fruit`, `edible_leaf`, `cuisine` presence)
    - indoors (`indoor`)
    - medicinal (`medicinal`)
    - poisonous
    - fruits/flowers presence
- Controlled vocabularies:
    - `plant_cycle`: *Perennial → Every year; Annual → Once a year; Biennial → Every 2 years*.
    - `growth_rate`: mapped to *Slow / Medium / Fast* where determinable.
    - `watering`: consolidated into a 5-bucket scale (*None / Minimum / Low / Average / Frequent*).
    - `sunlight` list values cleaned to *Full sun / Partial sun / Partial shade / Full shade* (case/variant normalisation, de-duplication, order preserved).
- **Multivalued attributes:** stored as **JSON arrays** in MySQL for flexible filtering and indexing

    - `sunlight`, `soil`, `attracts`, `propagation`, `pruning_month`, `pest_susceptibility`.
    - Empty lists are stored as `[ ]` (not NULL) to distinguish “known empty” from “missing”.
- **Care guide synthesis:** string fields prepared for user-facing display
    - `watering_general_benchmark` (e.g., *At least once {value} {unit}*)
    - `pruning_count` (e.g., *{amount} times {interval}*)
    - `flowers_detail` (e.g., *Flowers in {season}*)
- **Distribution maps:** captured as `full_url` (HTML) for rendering; stored as LONGTEXT.
- **Images:** `regular_url` for profile display; `thumbnail_url` for ML retrieval index building.

Data Quality Checks

- **Key integrity:** uniqueness of `plant_id` and `general_plant_id` within [1,3000]; no overlaps with threatened range.
- **Enum conformance:** values restricted to defined vocabularies; anomalies logged.
- **Boolean coherence:** e.g., edible TRUE requires supporting source evidence; otherwise flagged.
- **Name sanity:** scientific names checked for genus–species structure (non-fatal warnings).
- **Completeness:** presence of essential identity fields (IDs, names, core traits) monitored.

Outputs

- **Table01_PlantMainTable** (identity + high-value traits for search/filter).
- **Table02_GeneralPlantDescriptionTable** (type, cycle, attracts, propagation, description).
- **Table03_GeneralPlantCareGuideTable** (watering/sunlight/soil/pruning and care text).
- **Table04_GeneralPlantDistributionMapTable** (distribution link).
- **Table05_GeneralPlantImageTable** (regular + thumbnail URLs).

3.1.2. City of Melbourne - Threatened Plant Plan

Purpose

Provides conservation descriptors and simplified care context for Australian threatened plants. It enables biodiversity-aware views and complements Perenual traits on detail pages. It also distinguishes threatened vs general plants at identity level.

Extraction

CSV dataset downloaded from Melbourne Open Data. Source columns include conservation status, provenance, weed rating, habit, soil/sun guidelines, propagation methods/protocols, cultivation notes, and local benefits.

Transformation & Normalisation

- **ID assignment & merge strategy:** threatened plants are assigned `threatened_plant_id = 3001–3200` and **appended** into **Table01_PlantMainTable** to create a single identity table. For these rows, `general_plant_id = NULL`, `if_threatened = TRUE`.
- **Schema alignment:** column names and types harmonised with Perenual-derived tables to simplify joins and UI rendering (e.g., mapping `Sun` to `sun_expose` JSON list, applying the same cycle vocabulary via `Habit`).
- **Minimal booleans:** non-applicable flags (e.g., edibility) remain FALSE/NULL to avoid implying traits not present in source.

Data Quality Checks

- **Non-overlapping ID ranges** with Perenual.
- **Text normalisation** (trimming, consistent casing) for searchability.
- **Completeness** of key conservation fields; missing values logged.

Outputs

- **Table01_PlantMainTable** (threatened segment, identity and minimal traits)**.**
- **Table06_ThreatenedPlantDescriptionTable** (conservation_status, provenance, weed_rating, habit, benefits)
- **Table07_ThreatenedPlantCareGuideTable** (soil, sun, propagation, cultivation notes)

3.1.3. City of Melbourne – Urban Forest

Purpose

Provides spatial context of species distribution in Melbourne's urban canopy. Supports Iteration 2 insights and geospatial visualisation.

Extraction

CSV import with species names, dimensions, dates, and coordinates.

Transformation & Normalisation

- **Spatial typing:** `latitude`/`longitude` parsed into MySQL `POINT` (`coords`) with a **SPATIAL INDEX** to support efficient radius searches.
- **Type casting.** Numeric fields (e.g., diameters) cast to numeric; date fields normalised to DATE.
- **Taxonomic alignment.** Names trimmed and cased for consistent matching.

Data Quality Checks

- **Coordinate validity** and non-null checks.
- **Range checks** for numeric fields (e.g., plausible diameters).
- **Date sanity** (e.g., planted date <= last inspected)

Outputs

- **Table12_UrbanForestTable** (taxonomy, attributes, `coords`)

3.1.4. Open-Meteo – Weather Forecast

Purpose
Supplies short-term (16-day) weather outlook for location-aware recommendations (Iteration 2 insight). Enables comparing forecasted conditions with plant tolerances.

Extraction
On-demand API calls returning JSON forecasts by lat/long.

Transformation & Normalisation
- **Aggregation window.** Forecasts aggregated to daily values across a 16-day horizon for each query location.
- **Suitability inputs.** The required plant-side features (sunlight categories, watering needs, drought tolerance, minimum tolerable temperature) are prepared in **Table13_GeneralPlantListforRecommendation** from Perenual attributes; these are later compared with forecast summaries at runtime.

Data Quality Checks
- **Temporal completeness** across the horizon; gap handling.
- **Location coherence** (forecast coordinates match query).
- **Unit consistency** across metrics.

Outputs
- No permanent forecast table. Weather is consumed dynamically; the **plant-side features** reside in **Table13_GeneralPlantListforRecommendation**

3.1.5. TSX – Threatened Species Index

Purpose
Adds conservation trend context for Victoria to support Iteration 2 hindsight/foresight analysis.

Extraction
Annual CSV of index values per year (state level).

Transformation & Normalisation
- **Baseline rebasing.** Values normalised to 1985 = 1.0 for comparability.
- **Smoothing & uncertainty.** Generalised Additive Model (GAM) smoothing applied to reduce noise and handle gaps; bootstrap resampling used to derive 95% confidence bounds (`index_conf_low`, `index_conf_high`).
- **Export.** Retain post-1985 years; export compact four-column table.

Data Quality Checks
- Year coverage
- monotonic time order
- outlier inspection after smoothing

Outputs

- Feeds **Table14_TSX_Table_VIC**

3.1.6. Kaggle Plant Disease

Purpose
Provides labelled imagery to support disease recognition (Iteration 2 ML). Labels require harmonisation with Perenual's broader disease taxonomy.

Extraction
Image folders (≈19 GB) containing 91 labels in the form `<plant>__<disease>`.

Transformation & Normalisation

- **Label harmonisation.** Multi-step mapping from Kaggle labels to **Perenual disease IDs**: (1) exact name matches; (2) partial matches prioritised by host; (3) symptom-based similarity; (4) pest vector linkage; (5) "__diseased" mapped to the most common disease for that plant; (6) healthy mapped to ID 0.
- **Provenance.** Mapping notes retained for auditability.
- **Image selection.** Where multiple images are available per disease, at least one `regular_url` and `thumbnail_image` entry is recorded for profile vs model use.
-

Data Quality Checks

- Mapping coverage statistics
- conflict flags
- image integrity (readability, size > 0)

Output

- Feeds **Table09_PlantDiseaseTable**, **Table10_PlantDiseaseImageTable**, **Table11_PlantDiseaseLinkTable**.

3.1.7. TSX Enrichment with Climate Variables

Purpose
Enhance the Threatened Species Index (TSX) with annual climate covariates (temperature, precipitation, shortwave radiation) to support trend comparison and exogenous-variable forecasting (ARIMA/SARIMAX) in **Epic 7 (US 7.1, 7.2)**.

Sources
TSX annual index series (state and national, 2000–2021) and Open-Meteo historical weather (daily) aggregated to annual metrics (2000–2024); climate extended to 2027 via ARIMA-based forecasting.

Extraction

- Consolidated multiple TSX CSVs (states + national) into a harmonised dataframe (standardised column names, year as INT).

- Queried Open-Meteo by state-capital coordinates using the official Python client with caching/retries.

Transformation & Normalisation

- **Annual aggregation** from daily climate:
  - `annual_mean_temp` = mean daily temperature (°C)
  - `annual_precip_sum` = sum daily precipitation (mm)
  - `annual_radiation_sum` = sum daily shortwave radiation (MJ/m²)
- **National roll-up** as the mean of state metrics.
- **Temporal alignment**: merged TSX (2000–2021) with climate (2000–2024); corrected one boundary artefact (spurious 1999 row) and re-pulled missing slices via cache.
- **Forecast extension**: fit ARIMA per climate variable to forecast **2025–2027**; fit SARIMAX on TSX (2000–2021) with climate as exogenous (`exog`), then forecast **2022–2027** using climate forecasts

Data Quality Checks

- Unit consistency checks
- coverage audits (2000–2027 after model extension)
- residual diagnostics and parameter stability recorded in project notes

Output

A single, enriched annual table for visualising four trends (TSX, temperature, precipitation, radiation) with historical vs predicted flags, exported as `Table14_TSX_Table_VIC_version4.csv`.

3.1.8. State Shape Table

Purpose

Provide authoritative state/territory boundaries for choropleths, regional overlays, and spatial joins with TSX/forecast outputs in **Epic 7 (US 7.1, 7.2)**.

Sources

ASGS digital boundary files (Edition 3, 2021–2026).

Transformation & Normalisation

- Selected state-level layer; dissolved multipart geometries where applicable; stored as **MULTIPOLYGON** with **SRID 7844 (GDA2020)**.
- Normalised state names/codes to match the `state` values in Table14.
- Repaired minor topology issues to ensure `ST_IsValid = TRUE`.

Data Quality Checks

- CRS enforcement at ingest
- geometry validity checks
- duplicate name/code prevention.

Output

- `state` (TEXT, PK) and `geometry` (MULTIPOLYGON, SRID 7844) for map rendering and spatial joins

3.1.9. TSX Species Monitoring Table

Purpose
Store species-level monitoring attributes (taxonomy, conservation status, region centroids) to support drill-down, regional summaries, and future species-trend modelling in **Epic 6** and **Epic 7**.

Sources
TSX monitoring exports (species–year matrices and metadata).

Transformation & Normalisation
- Taxonomy normalisation (trim/case; consistent binomial formatting).
- Status fields retained (EPBC, IUCN); `MaxStatus` derived as the strictest category.
- Spatial typing: `RegionCentroidLatitude/Longitude` → MySQL `POINT coords` (SRID aligned to DB default) for proximity queries.
- `State`/`Region` values aligned to Table14 and StateShapeTable conventions.

Data Quality Checks
- Taxonomic uniqueness checks
- coordinate plausibility within AU bounds
- null audits for critical fields.

Output
- Wide table (~20 columns) grouped by taxonomy, conservation status, and location for region and species queries.

3.1.10. Australian Gardening Club

Purpose
Curated garden-club metadata to enable community discovery and engagement features in **Epic 8 (US 8.1–8.3).**

Sources
Garden Clubs of Australia listings and public club pages.

Transformations & Normalisation
- Schedule that was initially written as a string (e.g. “Every 2nd of Monday) parsed into `Meeting_day` (1–7), `Meeting_week` (1–5), `Meeting_hour` (local time text).
- URL validation and canonicalisation for `Link`; contact strings cleaned.
- `State`/`Location` standardised for filtering and optional geocoding.

Data Quality Checks
- Deduplication on (Club, State, Link)

- HTTP checks where accessible
- completeness audits for required fields (or, missing data is not relevant)

Output
- Final CSV with 182 rows, ready for application search/filter.

3.2. Created Dataset

Identity & Core

Table01_PlantMainTable — Unified plant identity for search/filter and detail pages
- **Key fields:** `plant_id` (PK; 1–3000 general, 3001–3200 threatened), `general_plant_id`, `threatened_plant_id`, names, core boolean flags (edible/indoors/medicinal/poisonous/fruits/flowers), `sun_expose` (JSON), `watering`, `plant_cycle`, `growth_rate`.
- **Upstream:** Perenual (species details), City of Melbourne (threatened plan).
- **Design rationale:** Single identity space simplifies joins and keeps UI logic uniform.
- PlantMainTable (Table 01): central identity table combining general plants (Perenual IDs 1–3000) and threatened plants (City of Melbourne IDs 3001–3200).

General-Plant Detail

Table02_GeneralPlantDescriptionTable — Descriptive attributes and narrative
- **Key fields:** `general_plant_id`, `if_*` flags, `plant_type`, `plant_cycle`, `attracts` (JSON), `propagation` (JSON), `description` (TEXT).
- **Upstream:** Perenual (species details).

Table03_GeneralPlantCareGuideTable — Care conditions and guides for rendering.
- **Key fields:** `general_plant_id`, `watering`, `watering_general_benchmark`, `sunlight` (JSON), `soil` (JSON), `drought_tolerant`, `salt_tolerant`, `pruning_month` (JSON), `pruning_count`, `pest_susceptibility` (JSON), `flowers_detail`, `harvest_season`, `growth_rate`, `maintenance`, `care_level`, and textual guides (`watering_guide`, `sunlight_guide`, `pruning_guide`).
- **Upstream:** Perenual (species details + care guides).

Table04_GeneralPlantDistributionMapTable — Distribution/hardiness map link
- **Key fields:** `general_plant_id`, `distribution_map` (LONGTEXT URL).
- **Upstream:** Perenual.

Table05_GeneralPlantImageTable — Profile and thumbnail images
- **Key fields:** `general_plant_id`, `regular_url_image`, `thumbnail_image`.
- **Upstream:** Perenual.

Threatened-Plant Detail

Table06_ThreatenedPlantDescriptionTable — Conservation descriptors & benefits.

- **Key fields:** `threatened_plant_id`, `conservation_status`, `provenance`, `weed_rating`, `habit`, `germplasm_source`, `conservation_benefit`, `local_benefits_description`, `horticultural_potential`.
- **Upstream:** City of Melbourne.

Table07_ThreatenedPlantCareGuideTable — Simplified care context
- **Key fields:** `threatened_plant_id`, `soil`, `sun`, `propagation_method`, `propagation_level`, `cultivation_note`.
- **Upstream:** City of Melbourne.

Disease Knowledge & Images

Table09_PlantDiseaseTable — Disease taxonomy and narrative.
- **Key fields:** `plant_disease_id`, `common_name`, `scientific_name`, `other_name` (JSON), `host` (JSON), `description` (JSON of sections), `solution` (JSON of sections).
- **Upstream:** Perenual.

Table10_PlantDiseaseImageTable — Disease image
- **Key fields:** `plant_disease_id`, `regular_url_image`, `thumbnail_image`.
- **Upstream:** Perenual (images), curated selection

Table11_PlantDiseaseLinkTable — Mapping audit (Kaggle → Perenual).
- **Key fields:** `kaggle_plant_disease_name`, `perenual_plant_disease_id`, `perenual_plant_disease_common_name`, `mapping_notes`.
- **Upstream:** Kaggle labels + internal mapping workflow + Perenual IDs.

Urban Forest (Spatial)

Table12_UrbanForestTable — City tree inventory with spatial indexing.
- **Key fields:** taxonomy, attributes, date fields, `latitude`, `longitude`, and `coords` (POINT) with a SPATIAL INDEX for efficient radius queries.
- **Upstream:** City of Melbourne (urban forest CSV).

Recommendation Support

Table13_GeneralPlantListforRecommendation — Plant-side features for suitability scoring.
- **Key fields:** `general_plant_id`, `plant_type`, `sunlight` (JSON), `watering`, `drought_tolerant`, `absolute_min_temp_c` (derived from Perenual `hardiness.min`).
- **Upstream:** Perenual attributes; used at runtime with Open-Meteo forecasts (no permanent forecast table).

Conservation Context

Table14_TSX_Table_VIC — Yearly threatened species trend for Victoria.

- **Purpose:** Unified historical TSX (2000–2021), annual climate covariates (2000–2024), and model forecasts (2022–2027 for TSX; 2025–2027 for climate) to support multi-trend visualisation and SARIMAX forecasting.
- **Key fields (7):**
  `year` (INT, 2000–2027),
  `index_value` (DOUBLE),
  `value_type` (TEXT: “Historical”/“Predicted”),
  `annual_mean_temp` (DOUBLE, **°C**),
  `annual_precip_sum` (DOUBLE, **mm**),
  `annual_radiation_sum` (DOUBLE, **MJ/m²**),
  `state` (TEXT).
- **Row count:** 168 × 7.
- **Notes:** Precipitation unit corrected to **mm**; baseline interpretation from earlier TSX processing (1985=1.0) retained conceptually, with available series beginning in 2000 for this table.

Table15_StateShapeTable

- **Purpose:** State/territory geometries for mapping and spatial joins with Table14 and other regional outputs.
- **Fields:** `state` (TEXT, PK), `geometry` (MULTIPOLYGON, SRID 7844).
- **Upstream:** ASGS boundaries (Edition 3).

Table16_TSX_SpeciesMonitoringTable

- **Purpose:** Species-level monitoring with taxonomy, conservation statuses, and regional centroids to enable regional drill-downs and future species-trend work.
- **Fields (groups):**
  Taxonomy: `ID`, `Binomial`, `CommonName`, `FamilyCommonName`, `Class`, `Order`, `Family`, `Genus`, `Species`, `Subspecies`, `FunctionalGroup`.
  Conservation: `EPBCStatus`, `IUCNStatus`, `MaxStatus`, `NationalPriorityTaxa`.
  Location: `State`, `Region`, `RegionCentroidLatitude`, `RegionCentroidLongitude`, `coords` (POINT).

<u>Community Based</u>

Table17_AustralianGardenClubTable – Community directory powering club discovery and scheduling features.

- **Fields:** `Club`, `Link`, `Meeting_day`, `Meeting_week`, `Meeting_hour`, `Contact`, `State`, `Location`.
- **Rows:** 182.

4. Data Storage

Database Engine

- We use MySQL to store all structured data in tables that match the ERD.

How data is saved

- CSV/API imports are turned into tables (Tables 01–17) with consistent column names and types.
- List/array fields from APIs (e.g., `sunlight`, `soil`, `attracts`, `propagation`, `pruning_month`, `pest_susceptibility`) are stored as MySQL JSON so we can keep their original list format.
- Images and maps are stored as URLs (TEXT/LONGTEXT). We do not store the actual image files in the database.

Spatial Data

- Where we have coordinates (e.g., UrbanForest and TSX region centroids), we also store a `coords` POINT column with a SPATIAL INDEX for fast “within radius” searches.
- State/Territory boundaries from ASGS are stored as `MULTIPOLYGON` with SRID 7844 (GDA2020) in Table15_StateShapeTable for accurate mapping and joins

Forecast flags

- Tables that mix history and predictions (e.g., Table14_TSX_Table_VIC) include a `value_type` field to label rows as “Historical” or “Predicted”

Loading into MySQL

- We import prepared CSV files using bulk load (as shown in our scripts) after we flatten/clean the API JSON to match the table schema and column order.

Indexes

- SPATIAL INDEX on `coords` where POINT geometry exists (e.g., UrbanForest), to speed up map and distance queries.
- BTREE indexes on primary keys and key identity columns (e.g., `plant_id`, `general_plant_id`, `threatened_plant_id`, `plant_disease_id`) to keep lookups and joins fast

Backups and Access

- Weekly full backups with test restores to make sure backups work.
- Role-based access so the app can read data safely and only approved users perform data loads/updates

Units and Consistency

- We keep the units as documented in the tables (e.g., temperature = °C, precipitation = mm, radiation = MJ/m²).
- Empty lists are stored as `[ ]` (not `NULL`) so we can tell the difference between “no values” and “missing data”.

5. Database Design

Core Hub-and-Spoke Model

- **Table01_PlantMainTable** is the hub for plant identity
  - **General plants:** `plant_id = general_plant_id` (1–3000).
  - **Threatened plants:** `plant_id = threatened_plant_id` (3001–3200), `general_plant_id = NULL`.
  - Holds high-value fields used by search/filter (names, edible/indoors/medicinal/poisonous/fruits/flowers, `sun_expose` JSON, `watering`, `plant_cycle`, `growth_rate`).
  - **Primary key:** `plant_id`

General-Series Spokes (1:1 optional from PlantMain)

- **Table02_GeneralPlantDescriptionTable** — FK: `general_plant_id → Table01.general_plant_id`.
- **Table03_GeneralPlantCareGuideTable** — FK: `general_plant_id → Table01.general_plant_id`.
- **Table04_GeneralPlantDistributionMapTable** — FK: `general_plant_id → Table01.general_plant_id`.
- **Table05_GeneralPlantImageTable** — FK: `general_plant_id → Table01.general_plant_id`.

Threatened-Series Spokes (1:1 optional from PlantMain)

- **Table06_ThreatenedPlantDescriptionTable — FK:** `threatened_plant_id → Table01.threatened_plant_id`.
- **Table07_ThreatenedPlantCareGuideTable — FK:** `threatened_plant_id → Table01.threatened_plant_id`.

Disease Module

- **Table09_PlantDiseaseTable** — disease identity and narrative (**PK:** `plant_disease_id`).
- **Table10_PlantDiseaseImageTable** — one image record per disease (optional) **FK:** `plant_disease_id → Table09.plant_disease_id`.
- **Table11_PlantDiseaseLinkTable** — maps Kaggle labels to Perenual IDs **PK:** `kaggle_plant_disease_name`; **FK:** `perenual_plant_disease_id → Table09.plant_disease_id`.
- Cardinalities: **Table09 → Table10:** 1:1 (optional); **Table09 → Table11:** 1:many.

Spatial & Regional

- **Table12_UrbanForestTable** — independent spatial inventory with `coords` (**POINT**) and a spatial index; not linked to PlantMain by FK.
- **Table15_StateShapeTable** — state/territory boundaries (**MULTIPOLYGON**, SRID 7844; **PK:** `state`).
- **Table16_TSX_SpeciesMonitoringTable** — taxonomy, status, region and centroid `coords` (**POINT**).

- Joins for mapping/aggregation: attribute join on `State` to **Table15**, or spatial containment (centroid within boundary).

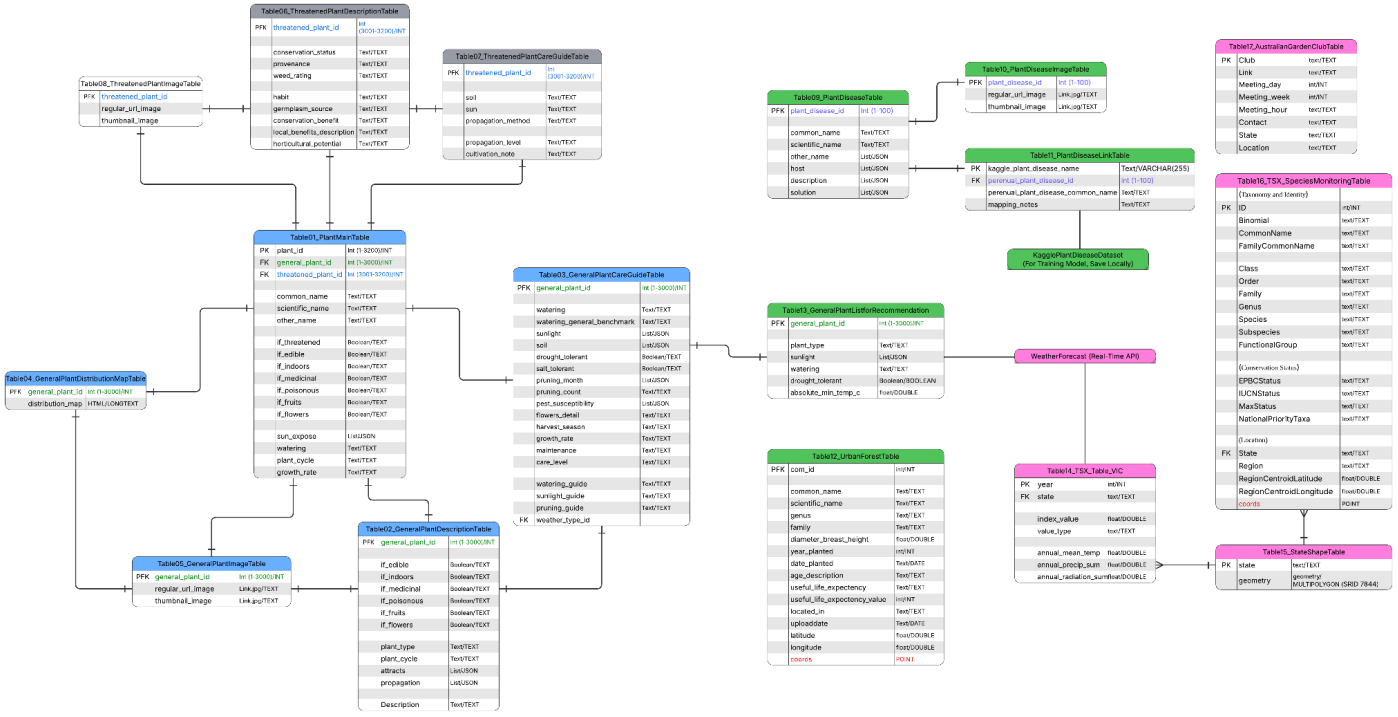
Time-Series & Recommendation

- **Table14_TSX_Table_VIC** — annual TSX values with climate covariates and `value_type` (“Historical”/“Predicted”).
  - Joins: `state` → `Table15.state`; `year` for time alignment.
- **Table13_GeneralPlantListforRecommendation** — plant-side features for suitability checks at runtime.
  - **FK:** `general_plant_id` → `Table01.general_plant_id` (1:1 optional).

Community Directory

- **Table17_AustralianGardenClubTable** — standalone lookup (Club, Link, meeting cadence, Contact, State, Location)
  - Filter-compatible with **Table15.state** (no FK).

ER Diagram: Link



6. Data Insights

6.1. Iteration 1

6.1.1. Hindsight

In Iteration 1 we combined general species from Perenual and threatened species from the City of Melbourne, so users could search everything in one place without thinking about sources. The single identity space in Table01_PlantMainTable made that possible (IDs 1–3000 for general; 3001–3200 for threatened).

From there, each page section pulled exactly what it needed from a focused table:

- **Table02_GeneralPlantDescriptionTable** supplies the "what is this plant" facts: type, cycle, propagation, and a readable description.
- **Table03_GeneralPlantCareGuideTable** carries the practical "how to care" details: watering, sunlight, soil, pruning hints, and small helper strings like *"At least once {value} {unit}."*
- **Table04_GeneralPlantDistributionMapTable** links straight to the species' hardiness/distribution map.
- **Table05_GeneralPlantImageTable** separates a clean profile image (for display) from a thumbnail (for ML), so the UI and the model never fight over the same asset.
- For threatened species, **Table06** and **Table07** add conservation status and simplified, local care context without breaking the unified search experience.

The key outcome: search, detail pages, images, and basic conservation context all worked together from day one and stayed simple to query.

6.1.2. Foresight

The way we shaped Iteration 1 data sets us up for smarter features later. Keeping sunlight/soil/propagation as lists (JSON) and normalising traits (e.g., `plant_cycle`, `watering`, `growth_rate`) means we can align plants with short-term weather conditions without reworking the base tables. The same identity split (general vs threatened) lets us layer biodiversity and conservation trends on top of the very same plant IDs. In short: the Iteration 1 schema is already ready for climate-aware recommendations and trend overlays introduced in later iterations.

6.1.3. Insight

How the features were actually built, table by table

- **Search & Filter (Epic 2 – US 2.1)**
  Primary filters (edible, medicinal, poisonous, fruiting, flowering) read straight from **Table01_PlantMainTable**. Sunlight and soil facets come from **Table03_GeneralPlantCareGuideTable** where available. Normalised terms (e.g., "Every year" for perennial) keep the UI language clear and consistent with the data.
- **Plant Profile (Epic 2 – US 2.2)**
  - **Identity & headline traits: Table01** (names, flags, watering, cycle, growth rate).
  - **Narrative & attributes: Table02** (type, propagation, attracts, description).

    - **Care sections: Table03** (watering/sunlight/soil/pruning, plus helper strings like `pruning_count` and `flowers_detail`).
    - **Map & images: Table04** (distribution link) and **Table05** (regular image for display).
    - **Threatened overlay:** when `if_threatened = TRUE` in **Table01**, the page also pulls **Table06** (status, provenance, benefits) and **Table07** (soil/sun/propagation/cultivation notes).
- **Why the split matters**
  Each table does one job well. The UI can render the essentials from **Table01** instantly, then add panels only when their data exists (e.g., show pruning only if **Table03** has pruning info). That keeps screens clean and performance predictable

6.1.4. Machine Learning Model (One-Shot Image Retrieval with CLIP)
We treat "plant recognition" as image retrieval rather than multi-class classification. Each species has (at most) one curated thumbnail (from Table05_GeneralPlantImageTable), so we build a gallery of class prototypes and return the most similar species for a user's query image. This design avoids heavy training, scales gracefully, and works with sparse data.

For more details, please refer to: Detailed Explanation on <Plant Image Recognition>.

Data & Artifacts

Gallery images (offline):
- Source: `thumbnail_image` URLs in **Table05** → downloaded to `01_raw_data/05_thumbnail_image/plant_species_thumbnail_image_{plant_id}.jpg`.
- Coverage: ~500–600 general plants with usable thumbnails (the set is expandable; threatened plants are out of scope for ML in Iteration 1).
- Identity join: `plant_id` keys back to **Table01_PlantMainTable** so the UI can open the correct profile and care panels

Index files (produced offline):
- `embeddings_fp16.npz`: matrix of L2-normalised prototype vectors (shape ≈ *N* × 512, float16).
- `meta.json`: model name (`openai/clip-vit-base-patch32`), embedding dim (512), build timestamp, counts, augmentation recipe hash, and a mapping of row index → `plant_id`.
- Optional: `thumb_manifest.csv` (plant_id, path, sha256) to audit gallery integrity.

Runtime cache:
- In-memory (or on-disk) copy of `embeddings_fp16.npz` and `meta.json` to avoid cold-start delays.
- Short-lived cache for recent query embeddings to support quick "re-rank" on UI toggles (e.g., crop vs. uncrop).

Offline Indexing Pipeline (One-time, Repeatable)
1. Load CLIP & Preprocessing

- Model: ViT-B/32 (512-d embeddings).
- Preprocess: CLIP's default resize/center-crop/normalise pipeline.

2. Data Quality Filter
    - Skip images that fail to open, <10 KB, or have extreme aspect ratios (to avoid logos/tiles).
    - Log and continue; do not block the run.
3. Augmentation Strategy → Class Prototype
    - For each gallery image: generate **20** light variants + 1 original (total **21**).
    - Typical ops (one per variant, small magnitudes):
        - Horizontal flip (p≈0.5)
        - Rotation (±5–10°)
        - Random crop (retain >= 90% area), then resize
        - Mild color jitter (brightness/contrast/saturation ±5–10%)
        - Small Gaussian blur (σ≈0.5) on a subset
    - Encode all 21 with CLIP → 21 vectors → **mean** → **L2 normalise** → **prototype**.
    - Rationale: the mean embedding is more stable to lighting/pose noise than a single pass.
4. Persist & Verify
    - Stack prototypes into `db_emb` (float16); save with aligned `plant_id` index.
    - Sanity checks: cosine self-similarity ≈ 1; no NaNs/Infs; row count equals number of accepted thumbnails.

Online Retrieval Pipeline (Per Query)

1. Preprocess & Test-Time Augmentation
    - Apply the same CLIP preprocessing to the uploaded image.
    - Create **7** light TTA variants (crop/rotate/color-jitter kept mild).
    - Encode 1 + 7 = **8** vectors; take the **mean** and L2 normalise → **query vector**.
2. Similarity Search
    - Compute **cosine similarity** via dot-product: `scores = db_emb @ query_vec`.
    - Return **Top-K** (default **K=5**) with `(plant_id, score)`.
3. Result Packaging
    - Join on **Table01** and **Table05** to populate the UI card: common_name, scientific_name, profile image, and quick traits (e.g., sunlight, watering).
    - The UI labels results as **"Most similar species"** (not "ground-truth ID") to set correct expectations.

Evaluation & Quality Control

We did not have a large, balanced, labelled photo set for formal accuracy reporting in Iteration 1, so we emphasised **procedural checks** and **scenario testing:**

- **Hold-out sanity set:** a small, hand-picked batch of photos scraped from public pages (when allowed) or captured by the team. We confirmed that the correct species (or a close cultivar) appears in Top-3 **most of the time** for well-lit, leaf-dominant images.
- **Ablation checks:**

    - Removing augmentations increases variance and produces more unstable Top-K.
    - Using float32 vs float16 embeddings showed **no practical difference** in rankings, so we kept float16 for memory.
- **Failure-mode tests:**
    - **Flowers only / extreme macro** → increased confusion across look-alike ornamentals.
    - **Background clutter** → minor effect; TTA + CLIP crop helped, but partial occlusion still reduces confidence.
    - **Very low light / heavy compression** → scores drop across the board; Top-K spreads out.
- **Integrity checks:** missing thumbnails, broken links, or duplicates are logged at build time (manifest cross-validated with **Table05**).

When we do gather a labelled test set, we'll report Top-1/Top-3 accuracy and "Hit@K" distributions in a later iteration; Iteration 1 focused on getting a reliable retrieval experience.

UX & Scoring Conventions

- **Scores:** cosine similarity in **[0, 1]** after L2 normalisation (we clamp negatives to 0 for display).
- **Confidence labelling:**
    - `score ≥ 0.30` → "High match"
    - `0.20–0.29` → "Likely match"
    - `< 0.20` → "Low confidence" (prompt user to try another angle or leaf shot)

      Thresholds were set empirically from our sanity set and are adjustable.
- **Top-K presentation:** show K=5 cards with names + profile image, then link through to the full plant page.

Reliability and Performance

- Latency:
    - Index load (warm): **~tens of ms** from memory.
    - Query encode: **tens to a few hundred ms** depending on CPU/GPU; matrix multiply is negligible at our current scale.
- Scalability:
    - Current exact search is fine for thousands to low-hundreds-of-thousands of classes.
    - If the gallery grows much larger, we can drop-in an ANN backend (e.g., Faiss IVF/HNSW) without changing the data contract.
- Rebuild triggers:
    - New thumbnails added to **Table05**
    - Augmentation policy change (hash mismatch in `meta.json`)
    - CLIP model/version change
- Reproducibility:
    - Deterministic seeds for augmentation; versioned `meta.json`; manifest with hashes for each gallery file.
    - Build logs capture counts, skips, and per-species vector norms.

Data Path Through the App

1. **Index build** uses **Table05** (thumbnails) and writes `embeddings_fp16.npz/meta.json`.
2. **Query** encodes the image → Top-K `plant_id`.
3. **UI join** uses **Table01** (names, flags), **Table02/03** (description & care), **Table05** (regular image) to render helpful next steps.
4. If the matched plant is threatened (`if_threatened = TRUE` in **Table01**), the detail page automatically surfaces **Table06/07** content for conservation context.

Known Edge Cases

- **No gallery image for a species** → species excluded from the index; not returned in results.

- **Multiple near-identical species** → Top-K shows several close options with similar scores; UI explains this as "visually similar species."
- **User-supplied non-plant images** → scores are low and diffuse; we display "Low confidence"

6.2. Iteration 2

6.2.1. Hindsight

Biodiversity trend context (Table14_TSX_Table_VIC)
By normalising TSX values to a 1985 baseline and retaining post-1985 series, we can show long-term directions of change rather than noisy year-to-year swings. This clarified which periods saw steeper declines and which stabilised shows a useful context to frame plant choices and conservation messaging in the app

Urban Forest distribution (Table12_UrbanForestTable)
After casting coordinates into a `POINT` and indexing them, radius queries surfaced clustering and dominance effects (e.g., pockets where a few species account for most plantings). That, in turn, highlighted **gaps** (rare/under-represented species) where diversification would help resilience.

Data alignment decisions
Using **Table01_PlantMainTable** as the single identity space let us join:
TSX (trend), Urban Forest (where species actually are), and Perenual traits (what a species *needs*). This made the browser-level stories coherent: a tree you see on the map has a profile, care guide, and (if threatened) a conservation context in one view.

6.2.2. Foresight

Better plant replacement
With Urban Forest spatial coverage (Table12) and trait filters from Perenual (Tables 02–03), councils and gardeners can shortlist candidates that **both** diversify the canopy **and** fit local conditions.

Short-term planning with forecast

Because **Table13_GeneralPlantListforRecommendation** holds comparable plant-side requirements (sunlight JSON, watering, drought tolerance, absolute minimum temperature), we can match these to **Open-Meteo** 16-day outlooks at runtime to suggest what is practical to plant **now** (Iteration 2 recommendation feature).

Clearer narratives for general public
Plotting TSX trends alongside plant stories gives users a "why it matters" line of sight: *this* plant supports pollinators, *that* group is trending down—here's how small choices help.

6.2.3. Insight

Map → Profile → Action flow
A user inspects a precinct on the Urban Forest map (Table12), taps a species, lands on the plant profile (Table01 + Table02/03 + Table05), and sees care guidance and distribution context (Table04). If a similar/related threatened plant exists, the page surfaces conservation info (Table06/07) to raise awareness.

Weather-aware suggestions
On a location query, the backend fetches the daily forecast and aggregates it. The rules engine compares those metrics to **Table13** and returns a list of candidates that won't fail in the next two weeks. Users don't get a generic list. They get **timely, locally viable** options

Consistent taxonomy across disease features
From **Table11_PlantDiseaseLinkTable**, the disease knowledge base (Table09/10) and the classifier output speak the **same ID language**. The UI can route predictions straight to the right disease profile and solution text.

6.2.4. Machine Learning Model

Plant Disease Recognition – Swin Transformer Classifier
Goal: Identify leaf diseases from images and map results to our standard IDs.

For more details, please refer to: Detailed Explanation on <Plant Disease Recognition>.

Training Data & Labels

- Images: Kaggle Plant Disease dataset (≈80k images, 91 raw labels).
- Label mapping: **Table11_PlantDiseaseLinkTable** reconciles Kaggle labels to **22** Perenual IDs (ID 0 = healthy), handling exact/partial matches, symptom similarity, and generic "__diseased" cases. This standardisation is what makes the classifier interoperable with **Table09/10**.

Model & Optimisation

- **Backbone:** Swin Transformer (`swin_base_patch4_window7_224`, ImageNet pre-trained).
- **Transfer learning:** Two stages—(1) freeze backbone, train head; (2) unfreeze and fine-tune end-to-end.
- **Class imbalance & regularisation: Focal Loss** to weight hard/rare classes; **Mixup/CutMix** to regularise; stratified 8:1:1 split.

- **Data loading:** Batched streaming via PyTorch `DataLoader` (avoids GPU OOM, keeps throughput high).

Evaluation & Reliability

- Best model reached **~98.3% test accuracy** across the 22 mapped classes.
- We return **Top-3** predictions with probabilities; the UI labels results as "Likely / Possible" and links to knowledge-base content (Table09).
- Known limitation: **closed-set** classifier. Out-of-scope diseases will still be mapped to the closest known class. (We mitigate this in the UI by showing confidence and encouraging a second photo if confidence is low.)

Output & Integration Path

- Artifacts: best weights (`.pth`) + `class_map.json` (model index ↔ Perenual ID).
- Backend translation: model index → Perenual ID → display name & solution (Table09), image (Table10).
- User impact: actionable guidance instead of raw labels; consistent terminology with the rest of the database.

Weather-Based Plant Recommendation – Rule-Based Engine

Goal: Recommend plants that can **survive and cope** with the **next 16 days** at a user's location.

For more details, please refer to: Detailed Explanation on <Plant Recommendation System>.

Prepared Feature Table

- `general_plant_id`, `plant_type`, `sunlight` (JSON), `watering`, `drought_tolerant`, `absolute_min_temp_c` (from Perenual hardiness.min).
- This is a compact, purpose-built slice for fast suitability checks.

Runtime Pipeline

- Forecast aggregation (Open-Meteo)
  - We compute: extreme min/max temperatures, average sunshine hours, average max UV index, average daily precipitation, average relative humidity over 16 days.
- Matching rules
  - **Temperature survival:** forecast **extreme_min_temp ≥ absolute_min_temp_c** (hard filter).
  - **Sunlight:** align sunshine duration & UV with the plant's `sunlight` categories (e.g., full sun ≥ ~6h; part shade ~3–6h; shade < ~3h), with UV sanity checks.
  - **Watering vs. precipitation:** filter obvious mismatches (e.g., "Frequent" watering vs. very dry forecast).
  - **Drought tolerance vs. humidity:** exclude non-tolerant plants under very dry air (e.g., <40% RH).
- Output:
  - shuffled list of suitable `general_plant_ids` + short weather summary for transparency.

Use of Rules
For a short horizon and well-understood botanical thresholds, a ruleset is faster, transparent, and tunable without retraining. It also matches how users reason ("too cold to plant this week") and keeps the explanation simple.

- The "Recommendations" panel shows **why** each plant qualified (e.g., "min temp safe; fits full sun; precip aligns with 'Average' watering").
- Clicking an item opens the Plant Profile (Table01/02/03/05/04) for care details.

6.3. Iteration 3

6.3.1. Hindsight

Where threatened plants have been observed
Historical monitoring records (taxonomy + region + state + region centroids) let us **reconstruct past distributions**. By joining `State/Region` with **Table15_StateShapeTable** (ASGS boundaries), we can render clean choropleths and point maps to show **which regions historically recorded particular threatened plants** and how coverage differs state-to-state.

How the overall data has shifted
The enriched TSX table combines **annual index values (2000–2021)** with **annual climate metrics** derived from Open-Meteo (mean temperature, precipitation, shortwave radiation). Smoothing the TSX series makes the **long-term direction** readable and comparable across states, while the aligned climate columns provide the context users need to interpret whether changes in the index coincide with warmer, drier, or brighter years.

A consistent spatial frame for comparison
Using ASGS state/territory boundaries (MULTIPOLYGON, SRID 7844) standardises mapping and ensures that TSX lines, monitoring points, and climate overlays align to the **same geography**, reducing ambiguity in cross-state comparisons.

6.3.2. Foresight

Short to medium-term biodiversity outlook
With **Table14** extended by time-series modelling, we present **predicted TSX values** alongside historical data. This gives a forward view for each region (National + states), helping users and stakeholders spot **downward trajectories** early.

Risk identification by region
Because the forecasts are stored per state with a `value_type` of **Historical**/**Predicted**, the UI can automatically surface **"watchlist" states** (e.g., sustained decline over the forecast window) and connect users to **species-level context** from **Table16** for local detail.

Cleaner climate context

Forecasted climate covariates (derived via ARIMA) sit next to TSX predictions, so users can see the **expected temperature/precipitation/radiation backdrop** when reading the biodiversity outlook. This keeps the narrative grounded without overstating causality.

6.3.3. Insight

Map → Trend
A user picks a state on the **ASGS map** (Table15). The system loads that state's **TSX time series with climate columns** (Table14) into a small multiple chart: historical line, forecast continuation, and mini spark-lines for temperature/precipitation/radiation. From there, a "View species context" link queries **Table16** to show **which threatened taxa are monitored in that state** and where records were historically concentrated.

Reading without statistics jargon
We avoid heavy statistical claims in the UI. Instead, we show **paired trend views** (TSX vs. climate) with plain-language captions (e.g., "Recent years have been warmer than the 2000s; the threatened species index has continued to trend down"). This reflects the modelling results while keeping interpretation **transparent and cautious**.

Consistent data across
The same **PlantMainTable** identity that powers Iteration 1–2 features remains the application's backbone; Iteration 3 adds an **analytical layer** (Tables 14–16) that explains **why conservation matters here and now** and what might happen next.

6.3.4. Machine Learning Model

TSX Enrichment and Forecasting (ARIMA + SARIMAX)

For more details, please refer to: Detailed Explanation on <TSX Trend Prediction**>.**

Data assembled

Modelling steps (per region)

1. Forecast climate first (ARIMA)
    - Separate ARIMA models per climate variable extend **2000–2024** to **2025–2027** (keeping forms simple where series behave like white noise).
2. Forecast TSX with exogenous inputs
    - TSX shows a **downward, non-stationary** pattern, so we use **SARIMAX(1,1,0)** with climate as exog to project **2022–2027**.
3. Label outputs
    - Concatenate historical and predicted rows and mark with `value_type = 'Historical' | 'Predicted'`. Save back to **Table14**.

Key reading of results

- The **AR(1)** term is highly significant, indicating **year-to-year momentum** in the index change.

- In this annual, macro view, **climate coefficients are not statistically significant**, so we present climate as **context**, not as causal proof. That stance is reflected in the product copy.

Artifacts produced

- Final flat file used to populate **Table14** (version 4).
- MySQL table ready for charting and choropleths (7 columns: `year, index_value, value_type, annual_mean_temp, annual_precip_sum, annual_radiation_sum, state`).

Backend API for Maps & Charts

A lightweight service helps in enriching tables and shapes for the front-end.

- **/api/map/geojson** → boundaries from **Table15** for base maps.
- **/api/map/data/{year}** → TSX values (Historical or Predicted) by state for that year from **Table14**.
- **/api/chart/data/{state}** → full 2000–2027 series for a state with climate columns, supporting line charts and annotations.

Species Monitoring Integration

Connect state-level trends to species-level context

- Taxonomy + status fields (EPBC/IUCN/MaxStatus) explain **which groups** are of concern.
- Region centroids (stored as `POINT`) enable **regional summaries** and map callouts beside the TSX panel.
- Naming and region conventions are aligned to **Table14** and **Table15**, so users can jump from a state trend to **relevant species and known locations** without friction.

How it helps users

- A **single place** to see how the threatened species trend has moved, what the **near-term outlook** is, and where **monitoring records** cluster.
- **Context without overclaiming**: climate trends are shown alongside the index so users can reason about potential relationships while the app remains careful about causality.
- Clear **next steps**: from a declining state trend, users can drill down to **species lists and locations** and link out to care/conservation actions already supported elsewhere in the system.